

Supplementary information S2 (box). Supplementary Methods and Table (related to Figure 2)

Description of domains shown in Figure 2

Domain	Description
RRM	RNA recognition motif, single-strand RNA (ssRNA)-binding
RG/RGG	RG/RGG box repeats are arginine glycine rich low complexity regions, may bind RNA or act as protein-protein interaction domains in shuttling
DEAD	DEAD and DEAH box helicase motif, unwinds RNA (and DNA)
zf-CCCH	Zinc finger motif type C-x8-C-x5-C-x3-H, ssRNA-binding
KH	KH-homology domain, ssRNA-binding
GTP_EFTU, GTP_EFTU_D2, GTP_EFTU_D3	GTP-elongation factor family, proteins usually consist of 3 structural domains, 2 oligonucleotide binding domains (D2 and D3) and a GTP-binding domain
dsrm	Double-stranded RNA binding motif
zf-CCHC	Zinc knuckle, C-x2-C-x4-H-x4-C, ssRNA-binding
LSM	Like Sm domain is found in snRNP complexes, bind A/U rich regions
OB_NTP_bind	Oligonucleotide/oligosaccharide-binding (OB)-fold, found in DEAD-box helicases in association with HA2 domain, regulates helicase activity through RNA binding
HA2	Helicase-associated domain, found in RNA helicases
G-patch	G-patch domain, ~48 amino acids with 6 conserved glycines, found in RBPs
IBN_N	Importin-beta N-terminal domain, RNA transport or RBP transport proteins
SAP	(SAF-A/B, Acinus and PIAS) motif, RNA/DNA-binding domain
TUDOR	Tudor domain, found in Tudor proteins, Tudor proteins are in complexes with RBPs
RnaseA	RNase A domain, ssRNA endonuclease
zf-C2H2_jaz	JAZ dsRNA-binding protein zinc-fingers, dsRNA-binding
MMR_HSR1	50S ribosome-binding GTPase domain, found in RBPs
KOW	KOW (Kyprides, Ouzounis, Woese) motif, found in a variety of ribosomal proteins
RNase_T	RNase T ssRNA exonuclease domain
MIF4G	MIF4G (Middle domain of eukaryotic initiation factor 4G (eIF4G)), RNA- (and DNA-) binding
zf-RanBP	RNA-binding Ran-binding-protein-like zinc finger
NTF2	Nuclear transport factor 2 (NTF2) domain, found in RNA export factors
PAZ	Piwi Argonaut and Zwilli (PAZ) domain, posttranscriptional silencing domain, binds siRNAs
RBM1CTR	C-terminal region found in hnRNPs
PAM2	PABP-interacting motif PAM2, found in RBPs
Xpo1	exportin 1 domain, RNA transport or RBP transport proteins
S1	S1 ssRNA-binding domain
HGTP_anticodon	Anticodon binding domain, found in aminoacyl-tRNA synthetases
tRNA-synt_2b	tRNA synthetase class II core domain (G, H, P, S and T), core catalytic domain of tRNA synthetases
Piwi	Piwi domain (P-element induced wimpy testis), posttranscriptional silencing domain, dsRNA guide hydrolysis of ssRNA
CSD	cold-shock domain, ssRNA/ssDNA binding
Ribosomal_L7Ae	domain found in ribosomal proteins L7Ae/L30e/S12e/Gadd45
RNase_Zc3h12a	ssRNA endonuclease domain found in Zc3h12a proteins, member of the NYN domain family
Anticodon_1	tRNA anticodon-binding domain, found in tRNA synthetases
R3H	R3H domain, R-x3-H conserved core, binds ssRNA/ssDNA

Methods

1. Selection of RNA-binding proteins (RBPs) and transcription factors (TFs)

To define RBPs, we searched the protein sequences of the human proteome, 105,237 isoforms in Ensembl database (release v75), with the domain-search algorithm *hmmer3.0*¹ for the presence of Pfam-defined protein domains (Pfam HMM profiles, release v27)², using a confidence cut-off with e-value <0.01. Next, we selected 799 Pfam domains (see supplementary tables) that are involved in RNA processes (either directly RNA-binding or processing or typically found in RNP complexes) and filtered out all isoforms containing the selected RNA-binding domains (RBDs). For the final gene list we took one representative protein isoform per gene, selecting either the isoform with the highest number of RBDs or, in cases of equal number of RBDs in all isoforms, the longest isoform. This procedure resulted in 2,103 putative RBPs. In a second filtering step we took out zinc finger proteins, which contained KRAB-, SCAN, and BTB domains exclusively found in DNA-binding proteins. We manually inspected the residual 1,962 proteins using information available on NCBI's Entrez Gene and publication records to further remove proteins, which were unlikely to be involved in RNA metabolism or presented pseudogenes and gene duplications. Finally, we manually added RBPs based on literature reports of the proteins or their orthologs, if they were missed by domain annotation, as some RBPs lack a Pfam-defined RBD. This resulted in a final census of 1,542 proteins.

For defining TFs, we downloaded the dataset from Vasqueraz *et al.*³ and matched the gene names and IDs to the current Ensembl v75 release, removing non-protein-coding genes and archived gene IDs. Of the 1,987 listed genes we traced 1,874 genes. We listed duplicated genes only once and removed all genes, which were also present in our curated RBP list (such as e.g. ZFP36 protein family), classifying these as TF misannotations. Our final list of TFs contains 1,704 proteins (see supplementary tables).

2. Expression analysis across 16 tissues

We used RNA-seq data from the human body map and aligned the raw reads with *tophat2* and *cufflinks2* (REFS 4,5) against the human genome (hg19) using default parameters. The human body map was provided by Illumina comprising single and paired-end sequencing data from polyA-selected RNA isolated from 16 adult human tissues (<http://www.ebi.ac.uk/arrayexpress/experiments/E-MTAB-513/>). We set a minimum expression threshold of $\text{rpkm}=1$ and compared $\log_2(\text{rpkm})$ values for RBPs, TFs, and the remaining protein-coding genes across the 16 tissues. For the cumulative abundance analysis of RBPs and TFs, we added rpkm values for both groups in each tissue. For the subdivision of mRNA-binding proteins (mRBPs) and ribosomal proteins we used our manually curated annotation categories (see next section). Downstream analysis was performed in R using *ggplot2*.

3. Categorization of RNA targets for RBPs

We manually inspected the list of 1,542 RBPs and defined categories for RNA targets based on literature reports as ribosomal proteins, diverse targets, mRNA-, tRNA-, rRNA-, snoRNA-, snRNA-, and ncRNA-binding. Proteins with reported RNA-binding properties but unknown natural targets, were grouped into an unknown targets category. All other noncoding RNAs were grouped together in the ncRNA-binding category, including miRNAs, piRNAs, 7SK, lncRNAs, MRP RNA and RNase P, 7SL, vtRNAs, Y RNAs, viral RNAs and telomerase RNA. RBPs binding DNA/RNA hybrids were categorized into the diverse category, together with RNA

nucleases and RNA exosome components. Proteins that are part of RNP complexes, but may not bind RNA directly, were assigned to their main RNA pathways, e.g. candidates involved in splicing or translation processes were classified as mRBPs.

4. RBP family definition, targets, and conservation analysis

We used Ensembl Compara⁶ to retrieve paralog information for the current Ensembl v75 proteome and formed protein families with an average 20% homology of the query and target percentage identity. We hand-curated cases where family relationships were not well defined and overlapped. We furthermore removed pseudogenes, read-through transcripts, cDNA clones, and open-reading frames, which had no protein-coding entry on NCBI's Entrez Gene. Using this approach, we defined 1,111 RBP families and 554 TF families. We defined target categories for RBP families by manually inspecting the annotation categories of the individual members in each family and assigning the common target group per family based on the most frequently found target group in the family.

For the conservation analysis of paralogous protein families, number of homologs and average homology to human were queried through Ensembl Compara for RBPs and TFs in 10 selected species (macaque, mouse, rat, xenopus, zebrafish, zebrafinch, chicken, fruitfly, *C. elegans*, and *S. cerevisiae*). We determined average homology for protein families in three subsequent steps. First, per homolog we took the average identity score of target and query protein. Second, if one gene had more than one homolog in the queried species we took the geometric average of the average scores of the homologs to arrive at a final average conservation score per gene. Third, per family we took geometric averages of the average conservation scores calculated for each individual member to give an average homology score per protein family. We grouped families into 4 different conservation categories with average scores of 0-20%, 20-40%, 40-60%, 60-85%, and 85-100%. The average conservation of RNA pathways was determined from the averages of the RBP families within target groups, which were conserved in *S. cerevisiae*.

5. Tissue specificity analysis of RBPs and RBP families

To assess tissue specificity of RBPs and TFs across human tissues, we used a microarray tissue atlas by Dezso *et al.*⁷ that profiled expression of 16,867 genes across 31 human tissues using the ABI Human Genome Survey Microarray platform version 2. We directly used the processed data deposited on GEO for further analysis, selecting per gene, if more than one probe was present, the probe with the highest reported minimum intensity value. We defined a tissue specificity score S that assesses the deviation from a uniform distribution as information content. S measures, analogous to the definition of sequence logos in information theory, the difference between the logarithm of the total number of tissues and the Shannon entropy of the expression values for a gene.

$$S = H_{max} - H_{obs} = \log_2(N) - (-\sum_{i=1}^N [p_i \times \log_2(p_i)])$$

where the relative frequency $p_i = \frac{x_i}{\sum_{i=1}^N x_i}$,

x_i is the microarray expression level for gene in tissue i , H_{max} =maximum possible entropy, H_{min} = observed entropy, N is the number of tissues, in this case 31.

The scale of the tissue specificity score S depends on the number of tissues, in the current dataset the maximum S (H_{max}) is reached when $S = \log_2(31) = 4.95$. The plots show the maximum expression value per gene against its tissue specificity score S . Assessing tissue

specificities of RBP families, we examined scores of the profiled genes per family and defined 4 categories of expression profiles for RBP families. In the first category we defined tissue-specific protein families where all members had tissue specificity scores $S \geq 1$. In the second category, we defined families where not all, but at least one member had a tissue specificity score $S \geq 1$, as families with tissue-specific members. We binned families into the third category, the gradient RBP families, if all their members had tissue specificity scores between $0.3 > S > 1$. In the final category of ubiquitous families, we grouped all protein families with tissue specificity scores for all their members $S < 0.3$.

6. Expression analysis of RBPs across developmental stages of human gonad and brain development

For the analysis of variation of expression of RBPs during human fetal ovarian development, we used a microarray study by Houmard *et al.*⁸, which profiled expression of 17,508 genes during different gestational weeks using the Affymetrix Human Genome U133 Plus 2.0 Array platform. We directly used the processed data made publically available for further analysis, selecting, if more than one probe per gene was present, the probe with the highest minimum expression value across the profiled stages. We used the geometric average for biological replicates of developmental stages. Tissue specificity scores were computed as described above for the developmental stages. We extracted the 1,461 RBPs profiled in this dataset and selected the top 200 RBPs with highest scores S , which showed the highest deviation from a uniform distribution across the developmental stages. In order to display dynamics of expression on a comparable scale we further normalized the microarray intensity values for each gene to fold changes over the mean of microarray intensities across the profiled stages (*Fold change* = $\frac{x_i}{\sum_{i=1}^N x_i} * 1/N$, where x_i =microarray intensity for stage i , N =number of stages). We used unsupervised clustering to present heatmaps of the $\log_2$ (fold change over mean).

For the analysis of RBP expression in brain, we downloaded the processed RNA-sequencing data for human brain development from the brainspan atlas (<http://www.brainspan.org>) and selected 12 stages of hippocampus development as model for assessing RBP expression changes during neural development. We used geometric average for biological replicates of developmental stages. To calculate tissue-specificity scores, we added pseudocounts of 0.1 to each rpkm value and computed tissue specificity scores as described above. We set a minimum expression cut-off for genes as the sum of rpkm values for a gene across the developmental stages $\sum_{i=1}^N (x_i) \geq 12$, where N =total number of profiled stages, and x_i = rpkm value in stage i . 1,402 RBPs were above this threshold. We selected the top 200 RBPs with highest tissue specificity scores S . As above, we further normalized rpkm values to fold changes over mean by dividing rpkm values over the mean rpkm across the developmental stages to compare expression changes at similar scale (*Fold change* = $\frac{x_i}{\sum_{i=1}^N x_i} * 1/N$, where x_i =rpkm for stages i , N =number of stages). We used unsupervised clustering to present heatmaps of the $\log_2$ (fold change over mean). Pearson correlation and heatmaps were generated using the gplots package in R.

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